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Blockchain-Based Farm-to-Market Traceability System

A COURSE LEVEL PROJECT REPORT

Submitted by

II-year Students of Bachelor of Technology

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in partial fulfillment of the course of

CSE223 / APPLIED BLOCKCHAIN

Academic Year 2026 – 2027 (Odd Semester)

DECLARATION

We affirm that the project work titled **"Blockchain-Based Farm-to-Market Traceability System"** being submitted in partial fulfilment for the award of the degree of **Bachelor of Technology** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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Date: 13-11-2026

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BONAFIDE CERTIFICATE

Certified that this project report **"Blockchain-Based Farm-to-Market Traceability System"** is the bonafide work of **"GURU PRASANTH S (9925009028), LEUN HAMMI D (99250040308), JEYAPRASANNA M (99250040310)"** who carried out the project work under my supervision.

Signature of supervisor

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Submitted for the Project Viva-voce / Review held at Kalasalingam Academy of Research and Education, Krishnankoil on 13-11-2026

Internal Examiner **External Examiner**

ACKNOWLEDGEMENT

We are thankful for the opportunity to carry out this project under the Skill Enhancement Course curriculum, which enhanced our engineering skills through hands-on application and encouraged us to develop impactful solutions aligned with the Sustainable Development Goals (SDGs).

We express our sincere gratitude to Late **Kalvivallal Thiru T. Kalasalingam** for his vision, Illayavallal **Dr. K. Sridharan, Chancellor**, and **Dr. S. Shasi Anand, Vice President**, Kalasalingam Academy of Research and Education (KARE), Tamil Nadu, India, for continuous encouragement in promoting innovation and societal impact.

We extend our heartfelt thanks to **Dr. S. Narayanan, Vice Chancellor**, for his leadership, for supporting the Skill Enhancement Course framework that bridges academic learning with real-world relevance.

Special thanks to our Project Supervisor, **Dr.P.Chinnasamy,** for constant guidance, encouragement, and expert insights throughout the project.

Lastly, we appreciate the support of our faculty members, non-teaching staff, friends, and family who motivated us throughout this experiential learning journey.

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PROJECT SUMMARY

Project Title	**Blockchain-Based Farm-to-Market Traceability System**
Project Team Members (Name with Register No)	**GURU PRASANTH S – 9925009028, LEUN HAMMI D- 99250040308, JEYAPRASANNA M- 99250040310**
Guide Name/Designation	**Dr.P.Chinnasamy – Associate Professor, KARE**
Program Concentration Area	Agriculture Technology (AgriTech) – Blockchain-Based Supply Chain Traceability and Transparent Price Discovery.
Technical Requirements	Frontend: HTML5, CSS3, JavaScript. Backend: Node.js, Express.js. Database: PostgreSQL. Blockchain: Solidity, Hardhat, Ethereum Sepolia Testnet. Wallet Integration: MetaMask, Ethers.js. Authentication: Google OAuth 2.0 and Email/Password Authentication. API Integration: REST APIs, AGMARKNET and data.gov.in. Development Tools: Visual Studio Code, Git, GitHub and npm. Hardware: Standard computer/laptop, smartphone and internet connectivity. Security: HTTPS, secure authentication, input validation and role-based access control. Traceability: Smart Contracts, Blockchain Records and QR Code Verification.

Engineering standards and realistic constraints in these areas		
Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic	Cost-effective web application development, affordable blockchain integration, reduced intermediaries, and transparent pricing for farmers.	✓
Environmental	Promotes sustainable agricultural supply chains, reduces paper-based documentation, and minimizes unnecessary transportation and distribution inefficiencies.	✓
Social	Improves farmers' access to market information, promotes fair pricing, builds consumer trust, and supports transparency among farmers, traders, and consumers.	✓
Ethical	Ensures transparent transactions, data integrity, user privacy, secure authentication, and prevents unauthorized modification of agricultural records.	✓
Health and Safety	Provides product traceability through recorded farming, harvesting, and distribution information, helping consumers make informed purchasing decisions.	✓
Manufacturability	Developed using web technologies and blockchain integration. Requires reliable hosting, database management, secure APIs, and compatibility across devices and browsers.	✓
Sustainability	Supports long-term digital agricultural record management, scalable blockchain-based traceability, reduced paperwork, and sustainable supply-chain practices.	✓

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LIST OF ABBERIVATION

S. No.	Abbreviation	Full Form
1	AgriTech	Agricultural Technology
2	API	Application Programming Interface
3	AI	Artificial Intelligence
4	DB	Database
5	GPS	Global Positioning System
6	HTML	HyperText Markup Language
7	CSS	Cascading Style Sheets
8	JS	JavaScript
9	IoT	Internet of Things
10	QR	Quick Response
11	UI	User Interface
12	UX	User Experience
13	URL	Uniform Resource Locator
14	HTTP	HyperText Transfer Protocol
15	HTTPS	HyperText Transfer Protocol Secure
16	JSON	JavaScript Object Notation
17	OAuth	Open Authorization
18	DBMS	Database Management System
19	SHA	Secure Hash Algorithm
20	P2P	Peer-to-Peer
21	DLT	Distributed Ledger Technology
22	OTP	One-Time Password
23	RFID	Radio Frequency Identification
24	UTC	Coordinated Universal Time

PROBLEM STATEMENT

Agriculture supply chains often face challenges such as lack of transparency, limited product traceability, price exploitation, and inefficient communication between farmers, sellers, and consumers. AgriTrace aims to provide a transparent and reliable digital agricultural ecosystem through blockchain-based traceability, direct market connectivity, and real-time information sharing.

ABSTRACT

Farm-to-market agricultural supply chains frequently suffer from fragmented records, intermediary-driven information asymmetry, weak price discovery, limited product provenance, and difficulty verifying quality and transport conditions. These limitations reduce consumer trust and can prevent farmers from receiving a fair share of the final product value. This research proposes AgriTrace, a blockchain-enabled farm-to-market traceability and supply-chain intelligence framework that combines batch-level digital identities, QR-based consumer access, Solidity smart contracts, role-based authorization, price provenance, quality verification, and transport events. The proposed architecture treats the QR code as a compact pointer to a persistent batch identity rather than as a storage container for the complete product history. Critical supply-chain events are recorded through smart contracts, while high-volume IoT observations can remain off-chain with cryptographic commitments anchored on-chain. The prototype can be implemented using Solidity, Remix IDE, an Ethereum-compatible test environment, MetaMask, JavaScript/ethers.js, HTML/CSS/JavaScript, QR-generation libraries, ESP32-based sensing hardware, GPS, and environmental sensors. A controlled simulation can evaluate 1,000–10,000 batches across multiple products and supply-chain participants. Performance can be measured using traceability completeness, tamper-detection rate, QR verification latency, IoT-to-blockchain delay, transaction throughput, gas consumption, scalability, and unauthorized-operation rejection. The principal research and patent-oriented contribution is the integration of a batch-centric identity and event graph that links origin, ownership, price transitions, quality verification, transport checkpoints, selected IoT evidence, and consumer verification within a common verifiable lifecycle. Keywords: agricultural traceability, blockchain, smart contracts, QR code, IoT, price provenance, supply-chain transparency, batch identity, event graph, food supply chain.

project aligned with Sustainable Development Goals (SDGs) no: SDG 2, SDG 8, SDG9, SDG 12, SDG 17

CHAPTER –I

INTRODUCTION

Agricultural products typically pass through several actors between production and final consumption. Farmers create the product, traders or aggregators consolidate and transfer batches, transporters move products between locations, retailers sell the products, cooperatives may coordinate producers, and consumers ultimately depend on available information to judge origin, quality, and authenticity. Because these activities are often recorded in separate systems, it can be difficult to reconstruct a trustworthy end-to-end history. AgriTrace addresses this fragmentation by representing each product batch through a persistent digital identity. The identity is connected to a sequence of trace events covering creation, ownership transfer, pricing, quality checks, transportation, and selected environmental measurements. A QR code provides a convenient humanfacing entry point: scanning the code retrieves the batch identifier and allows the frontend to query the blockchain-backed traceability record. Solidity smart contracts provide the core business logic. They enforce role-based permissions, validate state transitions, store critical batch metadata, maintain price and trace-event histories, and emit events for auditable changes. The frontend is responsible for user interaction, dashboards, QR generation and scanning, while IoT devices provide real-world measurements. This separation avoids treating blockchain as a replacement for every software component.

CHAPTER-II

LITERATURE REVIEW

Existing research demonstrates increasing interest in blockchain and IoT for agricultural traceability. GS1's traceability framework emphasizes identifying products and locations, capturing relevant data, and sharing information across critical tracking events and key data elements. FAO publications similarly identify distributed ledgers and smart contracts as promising mechanisms for transparency and traceability, while recognizing technical and institutional barriers. Caro et al. demonstrated a practical blockchain-based agricultural traceability architecture integrating IoT and blockchain technologies, with evaluations involving latency, CPU utilization, and network consumption. Subsequent studies explored Ethereum, Hyperledger Fabric, Hyperledger Sawtooth, RFID, trusted computing, decentralized identity, and enterprise blockchain interoperability. Systematic reviews have identified recurring challenges including scalability, interoperability, privacy, data authenticity, cost, regulatory compliance, and the dependence on off-chain information. Recent work increasingly combines blockchain with RFID and IoT and investigates evaluation frameworks for agricultural traceability. However, many approaches focus primarily on recording provenance rather than jointly modelling product identity, ownership, price evolution, quality evidence, transport checkpoints, and consumer verification as a persistent lifecycle graph. AgriTrace is positioned around this integration while using selective on-chain anchoring to control storage and transaction overhead.

Year	Method/Study	Tools/Technology	Contribution	Limitation
2018	Caro et al. – AgriBlockIoT	Ethereum, Hyperledger Sawtooth, IoT	Practical blockchain and IoT-based agricultural traceability.	Deployment and resource overhead; interoperability challenges.
2022	Zhang – IoT and Blockchain Traceability	IoT, Blockchain, Trusted Computing	Trusted agricultural traceability architecture.	Scalability and implementation complexity.
2022	Marchese & Tomarchio	Hyperledger Fabric, Smart Contracts	Role-based agrifood traceability	Enterprise deployment and interoperability issues.
2022	Agri-SCM-BIoT	Blockchain, IoT, Enterprise Blockchain	Interoperability-oriented supply chain model.	Privacy and scalability challenges.
2023	Lv et al. – Systematic Review	Blockchain Literature Analysis	Synthesized research on agricultural blockchain traceability.	Unresolved adoption and technical issues.
2023	Bosona & Gebresenbet – Blockchain Traceability Review	Blockchain Traceability	Analyzed trustworthy retrieval and traceability requirements.	Practical integration remains difficult.

2023	Hasan et al.	Fantom, Ethereum, Self-Sovereign Identity (SSI)	Peer-to-peer trusted traceability and identity management.	Complexity of decentralized identity and deployment.
2024	Pang et al. – Evaluation Framework	Blockchain Agricultural Traceability Evaluation	Evaluated agricultural blockchain traceability frameworks.	Standardization and participation barriers.
2024	Li et al.	Blockchain, RFID, MySQL, Optimized SM3	Multi-layer traceability and RFID integration.	Centralized detailed-data dependency and authenticity challenges.
2024	Pakseresht et al. – Systematic Review	Blockchain Literature Analysis	Identified blockchain characteristics important to agrifood systems.	Technical, environmental, and adoption issues.
2024	Costopoulou et al.	Blockchain, IoT	Studied food traceability in olive-oil supply chains.	Domain-specific deployment considerations.
2024	Blockchain and Smart Contracts Study	Ethereum, Remix IDE	Automated product origin, quality, and	Prototype-level limitations.

			traceability processes.	
2025	Blockchain and IoT Smart-Farming Review	Blockchain, IoT	Reviewed applications, benefits, and challenges in smart farming.	Scalability, interoperability, and adoption remain open challenges.

CHAPTER III

PROBLEM DEFINITION AND PROJECT OBJECTIVES

3.1 Problem Definition

Agriculture plays an important role in the Indian economy, but farmers still face several challenges when selling their agricultural products. One of the major problems is the involvement of multiple intermediaries between farmers and final buyers. This can reduce the profit received by farmers and make the selling process less transparent.

Farmers may also find it difficult to identify suitable buyers, compare market prices across different districts and understand where their products can be sold at a better price. In addition, information about the origin, harvesting date, storage conditions and transportation history of agricultural products is often maintained through disconnected or traditional methods.

These limitations can affect trust between farmers, sellers and consumers. Buyers may not have sufficient information to verify the source and handling history of the products they purchase.

To address these challenges, AgriTrace proposes a digital farm-to-market platform that connects farmers and sellers, supports transparent price discovery and provides product traceability using blockchain technology.

The platform aims to make agricultural transactions more accessible, transparent and reliable through a combination of web technologies, blockchain and IoT-based monitoring.

3.2 Project Objectives

The main objectives of AgriTrace are:

1. To develop a digital platform that connects farmers directly with sellers and buyers.

2. To reduce unnecessary dependency on intermediaries by providing a common marketplace.

3. To enable farmers to list their agricultural products, including fruits, vegetables and other farm produce.

4. To provide a transparent system for product pricing, offers and order management.

5. To support district-wise agricultural market price comparison so farmers can make better selling decisions.

6. To maintain product origin, harvesting and supply-chain information through batch-level traceability.

7. To use blockchain technology to improve the integrity and verifiability of important transaction and traceability records.

8. To provide QR-based product verification for buyers and consumers.

9. To support IoT sensor integration for monitoring storage and transportation conditions.

10. To provide separate Farmer and Seller portals with role-based authentication.

11. To introduce AgriTrace Points as a reward mechanism for eligible platform activities.

12. To create a user-friendly and accessible platform that can support farmers with different levels of digital experience.

3.3 Proposed Solution

AgriTrace provides a centralized digital interface where farmers can register, list their products and interact with sellers. Sellers can browse available products, communicate offers and place orders.

Each agricultural batch can be associated with a unique identification number. Relevant information about the product journey can be recorded and made available through QR-based verification. Blockchain technology can be used to preserve the integrity of selected records.

The platform also aims to integrate official agricultural market information and IoT-based monitoring to provide additional value to farmers and buyers.

The overall objective is to create a more transparent and connected agricultural supply chain.

CHAPTER- IV

Proposed Methodology

4.1 System Architecture Overview

The proposed system follows a layered architecture consisting of the user layer, application layer, blockchain layer, IoT/data layer, and verification layer. The user layer contains farmers, traders, transporters, retailers, cooperatives and consumers. The application layer provides dashboards, batch registration, transfer operations, quality recording, price history, QR generation and consumer trace pages. The blockchain layer contains Solidity smart contracts. The contracts maintain user roles, batch identifiers, lifecycle states, ownership transfers, trace events, price history and selected verification records. The IoT layer contains devices such as ESP32, temperature and humidity sensors and GPS modules. The verification layer links QR scans to batch identifiers and retrieves verifiable history. Recommended high-level flow: Farmer → Create Batch → MetaMask Transaction → Smart Contract → Batch ID → QR Generation → Consumer Scan → Traceability Web Page → Blockchain Query → Origin / Price / Quality / Transport / IoT Evidence → Verification.

4.2 System Architecture



4.3 Batch-Centric Digital Identity

Each agricultural batch receives a unique identifier. The QR code should contain only a compact reference such as a batch ID or a URL containing the batch ID. It should not contain the complete blockchain history. This keeps the QR payload small and permits the underlying record to evolve while preserving the same public identity.

4.4 Smart Contract Functions

- registerUser() – registers an actor in the system.
- assignRole() – assigns or validates farmer, trader, transporter, retailer or verifier roles.
- createBatch() – creates a new agricultural batch and establishes its initial identity.
- getBatch() – retrieves core batch metadata.
- addTraceEvent() – records a lifecycle event.
- getTraceHistory() – retrieves the batch event history.
- transferBatch() – changes authorized batch ownership/custody.
- recordPrice() – records a price transition with timestamp and actor.
- getPriceHistory() – retrieves price provenance.
- verifyQuality() – records quality verification information.
- addTransportRecord() – records a transport checkpoint.

4.5 QR-Based Verification

For example, a QR payload may point to a trace page containing a batch identifier. The consumer scan opens the page, the frontend extracts the batch ID, and the application queries the smart contract. The consumer can then inspect origin, custody, recorded prices, quality checks, transport events and selected IoT evidence.

4.6 Selective IoT Anchoring

High-frequency sensor data should not automatically be written as individual blockchain transactions because this can increase cost and reduce scalability. Instead, raw measurements may be retained off-chain, while selected checkpoints,

summaries, hashes, or integrity commitments are anchored on-chain. This creates a practical balance between evidence integrity and blockchain efficiency.

4.7 Patent-Oriented Technical Positioning

The patent-oriented contribution should not be stated simply as 'blockchain + QR + IoT', because these individual technologies and their combinations have substantial prior art. A stronger technical focus is the specific interaction mechanism: a batch-centric agricultural digital identity is persistently bound to ownership transitions, price transitions, quality verification, transport checkpoints, selected IoT evidence, and consumer verification through a cryptographically linked event graph, with selective IoT anchoring used to reduce on-chain storage and transaction overhead. Potential claim categories include a system claim covering the architecture and data relationships, a method claim covering creation and verification of the batch-event graph, and technical-effect claims relating to tamper detection, traceability completeness, reduced on-chain data volume, and efficient verification. Patentability cannot be guaranteed; a formal prior-art and patent search should be completed before filing.

CHAPTER V

REQUIREMENTS AND MODULE DESCRIPTION

5.1 Hardware Requirements

The following hardware components are considered for the proposed IoT monitoring system.

|Component | Purpose |

|---|---|

| ESP32 Development Board | Main controller for collecting and transmitting sensor readings |

| DHT22 / SHT31 / BME280 | Temperature and humidity monitoring |

| DS18B20 Temperature Sensor | Optional temperature measurement at a specific storage location |

| GPS Module (Optional) | Location tracking during transportation |

| OLED Display (Optional) | Displaying local sensor readings |

| Buzzer (Optional) | Local warning indication |

| Power Supply / Battery | Providing power to the monitoring device |

| Connecting Wires and PCB | Electrical connections and circuit integration |

| Protective Enclosure | Protecting electronics during operation |

The final sensor selection depends on the prototype design, budget and accuracy requirements. These components should not be considered physically integrated until the hardware assembly and testing are completed.

5.2 Software Requirements

The software technologies used or planned for the AgriTrace application include:

- HTML5: Provides the structure of the web application.

- CSS3: Controls the visual appearance, layout and responsive design.

-JavaScript: Handles user interactions, frontend logic and API communication.

- Node.js: Provides the server-side JavaScript runtime.

- Express.js: Can be used to develop backend APIs.

- PostgreSQL: Stores user information, product listings, orders and traceability records.

- Solidity: Used to develop blockchain smart contracts.

- Hardhat: Supports smart-contract development, compilation and testing.

-ethers.js: Connects the application with blockchain networks and smart contracts.

- MetaMask: Provides browser-wallet connectivity.

-Google OAuth / Google Identity Services: Supports Google-based user authentication.

- GitHub: Supports source-code management and project hosting.

5.3 Functional Requirements

The application is expected to support the following functions:

1. Farmer and Seller registration and login.

2. Google authentication along with email-based authentication.

3. Role-based access to the appropriate dashboard.

4. Product listing and product browsing.

5. Product search and category filtering.

6. Offer submission and order management.

7. Agricultural market price comparison.

8. QR-based batch traceability.

9. Blockchain record verification.

10. AgriTrace Points management.

11. IoT sensor data collection and monitoring.

12. User-friendly language and voice-support features.

5.4 Module Description

Module 1: User Authentication

This module provides separate registration and login facilities for Farmers and Sellers. Users can access the platform through email-based authentication or Google authentication.

The system is designed to associate one registered email address with one account role, preventing the same email from being independently registered as both a Farmer and a Seller.

Module 2: Farmer Portal

The Farmer Portal allows farmers to manage their agricultural activities through a single dashboard.

Its functions include:

- Creating and managing product listings.

- Adding product images and descriptions.

- Entering available quantity and expected price.

- Viewing incoming offers and orders.

- Managing order-related activities.

- Accessing AgriTrace Points.

- Viewing relevant agricultural market information.

Module 3: Seller Portal

The Seller Portal allows sellers to explore available agricultural products and interact with farmers.

Its functions include:

- Browsing available products.

- Searching by crop and category.

- Viewing product quantity and pricing.

- Submitting purchase offers.

- Managing orders.

- Tracking order status.

- Viewing reward-point information.

Module 4: Agricultural Marketplace

This module acts as the connection between farmers and sellers.

Farmers can publish their available products, while sellers can discover and select products according to their requirements. The marketplace is intended to reduce communication gaps and make agricultural trading more organized.

Module 5: Tamil Nadu Agriculture Datasets

This module is intended to provide access to agricultural datasets and crop-related information.

It can include:

- Crop names and classifications.

- Seasonal growing and harvesting information.

- Average market prices.

- Crop storage and handling information.

- Shelf-life and spoilage-related guidance.

- Official Tamil Nadu Government agricultural resources.

Official government sources are important for maintaining the reliability of agricultural information. Market prices should display their source and observation date. Dataset links should be clearly distinguished from data that is actually retrieved and updated within AgriTrace.

Module 6: District-Wise Market Price Comparison

This module is designed to help farmers compare agricultural prices across different districts and markets in Tamil Nadu.

For example, a farmer selecting Tomato may compare available records from Salem, Madurai, Coimbatore, Erode and other supported markets.

The module can display:

- Minimum price.

- Maximum price.

- Modal price.

- Market or mandi name.

- District name.

- Price unit.

- Observation date.

- Original data source.

The comparison should use the latest common available observation date where possible, while also showing the individual observation date of every market record.

Module 7: Blockchain-Based Traceability

This module supports the recording and verification of important agricultural product information.

Each produce batch can be assigned a unique Lot ID. Relevant events such as product registration, harvesting, packing and dispatch can be associated with that batch.

Selected records can be represented by cryptographic hashes and stored on a blockchain. This allows users to verify whether the recorded information has been modified after being registered.

Detailed records and private user information should remain in the database rather than being directly published on the blockchain.

Module 8: QR Code Verification

A QR code can be generated for each supported produce batch.

When a buyer or consumer scans the code, the system can display the available product information, including:

- Product name.

- Batch identification number.

- Origin information.

- Harvest details.

- Available supply-chain events.

- Verification status.

This provides a simple way to access product information without manually searching through the application.

Module 9: IoT-Based Monitoring

The IoT module is designed to connect an ESP32 device with suitable temperature and humidity sensors.

The sensors collect environmental readings from storage or transportation environments. The ESP32 can transmit these readings to the backend through a network connection.

The dashboard can then display the readings, timestamps, device status and warning notifications.

This module is intended to improve visibility into storage and transportation conditions. Actual product freshness or spoilage prediction would require crop-specific validation and should not be inferred from temperature and humidity alone.

Module 10: AgriTrace Points

AgriTrace Points is a proposed reward system that encourages eligible platform activities.

Points can be awarded based on configured rules, such as daily participation or successful order completion.

The system should maintain a transaction history so users can understand when and why points were added. Duplicate rewards must be prevented through backend validation.

CHAPTER VI

SYSTEM IMPLEMENTATION

AgriTrace-GitHub

6.1 Implementation Overview

The AgriTrace application is designed as a web-based agricultural platform using frontend, backend, database and blockchain technologies.

The implementation follows a modular approach so that each major function can be developed and maintained independently while communicating with the other modules.

The frontend provides the interface through which farmers and sellers interact with the system. The backend manages authentication, business logic and data processing. PostgreSQL is intended to maintain application records, while blockchain technology supports the verification of selected records.

6.2 Frontend Implementation

The frontend is developed using HTML, CSS and JavaScript.

HTML defines the page structure, CSS provides the visual design and JavaScript manages interactive features.

The frontend includes:

- Home and project introduction page.

- Farmer and Seller authentication pages.

- Role-specific dashboards.

- Agricultural product marketplace.

- Dataset and agricultural information section.

- Market price comparison interface.

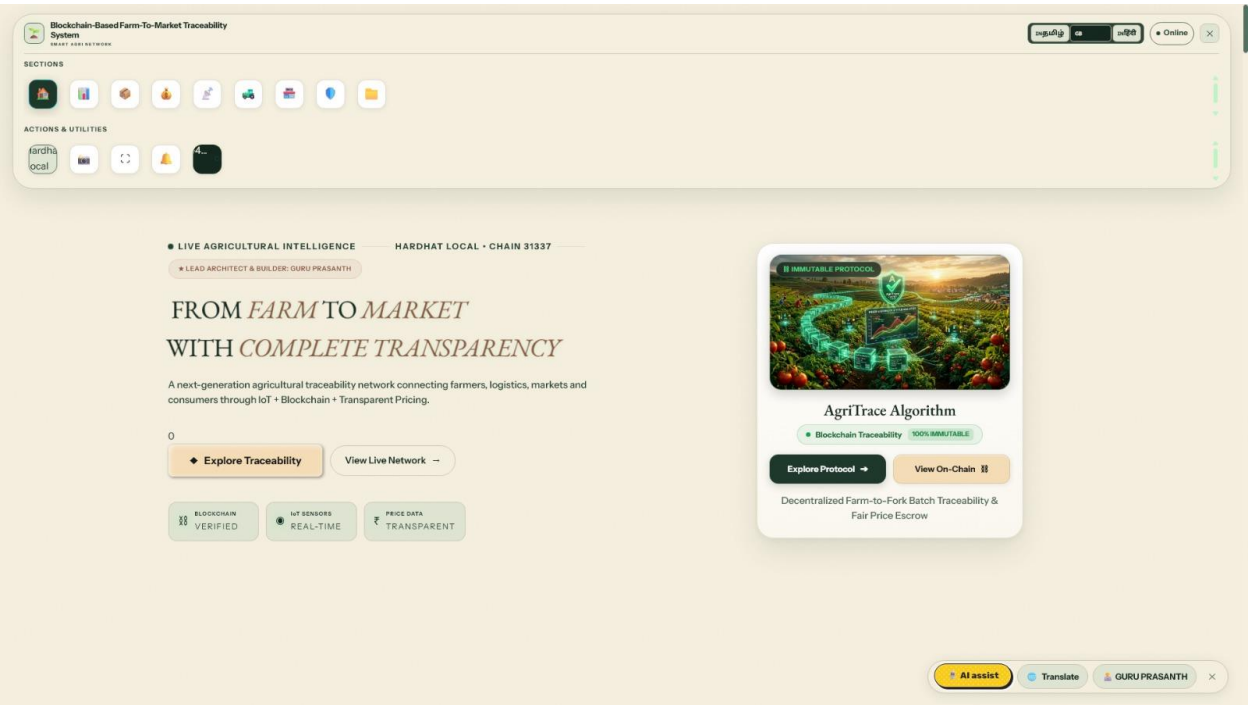
- QR-based traceability interface.

- Voice and language-support interfaces.

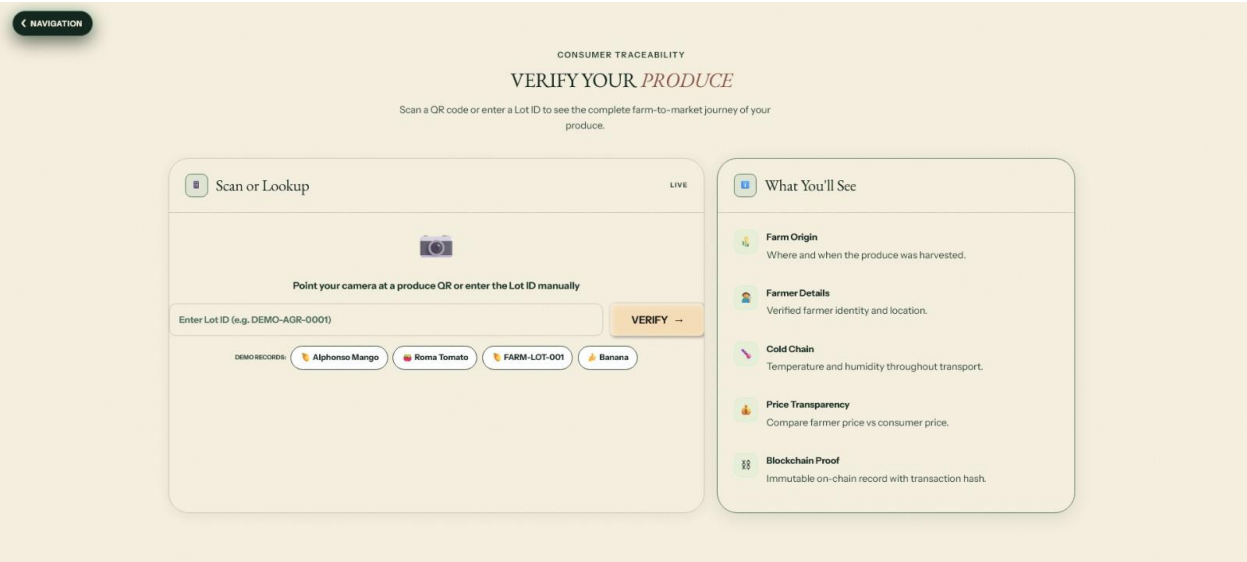
- AgriTrace Points section.

The interface is designed to provide a clear navigation experience and support different screen sizes.

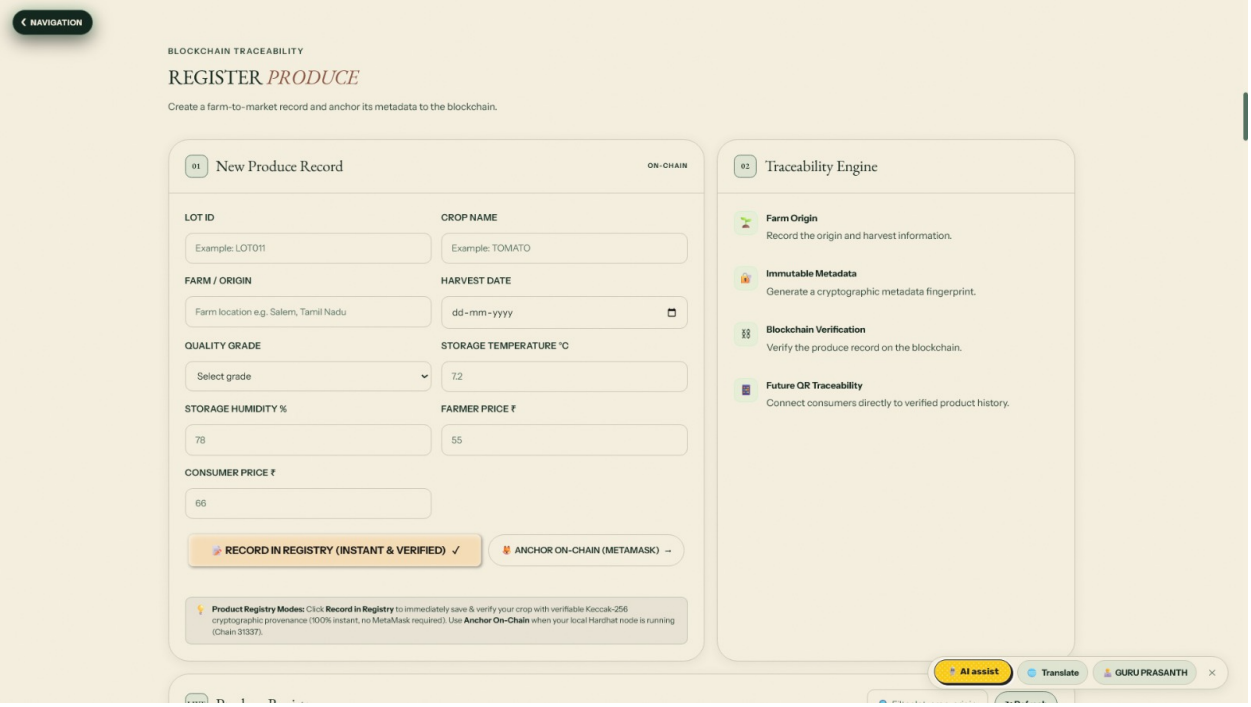
Home page



Verification Page



Product Registration Page



6.3 Backend Implementation

The backend is responsible for processing requests from the frontend and communicating with the database and other services.

The backend is designed to handle:

- User registration and login.

- Google authentication verification.

- Role-based access control.

- Product listing management.

- Offer and order processing.

- Batch and traceability information.

- Market price data retrieval.

- Reward-point calculations.

- Sensor data ingestion.

- Blockchain-related operations.

Server-side validation is important because frontend validation alone cannot protect sensitive application operations.

6.4 Database Implementation

PostgreSQL is used as the intended relational database technology for maintaining structured application information.

The database design can include separate records for users, Farmer and Seller profiles, products, offers, orders, batches, traceability events, sensor readings and reward transactions.

Relationships between these records help maintain consistency and make information retrieval easier.

For example, a product listing can be associated with a farmer, while an order can connect a seller with the selected product listing.

6.5 Google Authentication Implementation

Google authentication provides users with an alternative way to access AgriTrace.

The Google Identity Services flow allows users to authenticate using their Google accounts. The backend must verify the received identity token before creating an authenticated application session.

The system also checks the registered role of the email address to maintain the separation between Farmer and Seller accounts.

Sensitive credentials must be maintained securely on the backend and must not be exposed in frontend source files.

6.6 Blockchain Implementation

Blockchain functionality is designed using Solidity smart contracts and the Hardhat development environment.

The smart contract can store selected verification information, such as a batch-event hash or transaction reference.

The application can connect to the blockchain through ethers.js and a compatible wallet such as MetaMask.

When a user performs a blockchain-related operation, the application should display the actual transaction status and verification information.

6.7 QR Code Implementation

QR-based traceability provides a convenient method for accessing batch information.

A QR code is associated with a unique batch identifier or a safe public lookup URL. When scanned, the application retrieves the corresponding batch information from the backend.

Where blockchain verification is implemented, the application can also compare the stored record with its corresponding blockchain proof.

6.8 IoT Implementation

The proposed IoT implementation uses an ESP32 controller connected to temperature and humidity sensors.

The process follows these steps:

1. The sensor measures the surrounding environmental conditions.

2. The ESP32 reads the sensor values.

3. The device attaches the relevant device and timestamp information.

4. The readings are transmitted to the backend.

5. The backend validates and stores the readings.

6. The dashboard displays the received values.

7. Alerts can be generated when configured thresholds are crossed or when data becomes stale.

This approach creates a connection between physical agricultural storage conditions and the digital traceability platform.

6.9 Integration and Testing

The modules are intended to work together through defined interfaces. The frontend communicates with the backend using APIs. The backend communicates with PostgreSQL and, where configured, blockchain services and IoT devices.

Testing should cover user authentication, role restrictions, product listing, order processing, QR verification, blockchain transactions and sensor-data transmission.

Only features that pass the relevant tests should be presented as fully operational.

CHAPTER VII

RESULTS AND DISCUSSION

7.1 Results

AgriTrace presents a digital approach to improving communication and information sharing within the agricultural supply chain.

The developed web interface brings several agricultural functions into one platform, allowing users to explore the proposed marketplace, traceability and agricultural information features.

The main expected outcomes include:

1. A common digital platform for farmers and sellers.

2. A more organized product-listing and discovery process.

3. Improved visibility of product details and transaction information.

4. A structured approach to recording agricultural batch information.

5. A QR-based method for accessing product traceability details.

6. A foundation for blockchain-based record verification.

7. A framework for district-wise agricultural market price comparison.

8. A foundation for IoT-based environmental monitoring.

9. A reward mechanism through AgriTrace Points.

10. A user-friendly interface that can be extended with additional agricultural services.

7.2 Performance Comparison

1. Price Discovery

In traditional trading, farmers may need to contact different buyers or visit multiple markets to understand the available prices. AgriTrace aims to simplify this process by presenting market-price information in one place. District-wise comparisons can help farmers identify price variations between markets and make more informed selling decisions.

2. Product Traceability

Traditional product records may be maintained manually, making it difficult to verify the complete history of a product batch. AgriTrace supports digital batch records and blockchain-based verification for selected information. This can improve record integrity and make available traceability details easier to access.

3. Communication and Market Access

Traditional agricultural trading often depends on personal contacts and intermediary networks. AgriTrace provides a digital platform where farmers can list products and sellers can discover available produce. This creates an additional channel for communication and potential direct trading.

4. Data Management

Manual records can be difficult to update, organize and retrieve. AgriTrace uses a centralized database to manage users, products, listings, orders and other application information. This helps organize information and makes it easier to retrieve when required.

5. Storage and Transportation Monitoring

Traditional monitoring may depend on periodic manual checks. AgriTrace is designed to support IoT integration through sensors connected to an ESP32. Once implemented and tested, this feature can help users observe environmental conditions during storage and transportation.

7.2.4 Expected Performance Benefits

The proposed system is expected to provide the following improvements:

- Better price awareness: Farmers can compare available market prices before deciding where to sell their produce.

- Improved traceability: Digital batch records and blockchain verification can strengthen confidence in recorded product information.

- Reduced information gaps: Farmers and sellers can access relevant product and market information through one platform.

- Organized record management: Digital storage makes product, user and transaction records easier to manage.

- Improved accessibility: Users can access the platform through supported web browsers and devices.

- Future monitoring capability: IoT integration can provide additional visibility into storage and transportation conditions.

7.2.5 Performance Evaluation Parameters

To evaluate the actual performance of AgriTrace, the following parameters can be measured during testing.

Parameter What it measures Evaluation method

Page loading time Time taken for a page to become usable Measure page load duration

Login response time:

Time taken to complete authentication

Record response time over repeated tests

Product listing response:

Time taken to save and display a listing

Measure API response and database operation

Price retrieval time:

Time required to display market-price information

Test with the same dataset and network

QR verification time:

Time required to retrieve a batch record

Measure from scan/request to result

Blockchain transaction time:

Time required for a transaction to be confirmed

Record submission and confirmation timestamps

Sensor update interval:

Frequency of IoT readings received by the backend

Compare sensor and server timestamps

System availability:

Whether the platform is accessible during the test period

Monitor successful and failed requests

Note: Actual numerical performance values should be added only after conducting tests. Do not report estimated values as measured results.

7.2.6 Overall Discussion

The comparison indicates that AgriTrace offers a more organized and digitally connected approach to agricultural trading than conventional methods that depend heavily on manual communication and record-keeping. Its main contribution is the combination of a farmer–seller marketplace, market-price comparison and digital product traceability in one platform.

However, the effectiveness of the system depends on the accuracy of the information entered, the availability of reliable market-price data, successful blockchain integration and proper testing of the IoT hardware. Therefore, AgriTrace should be considered a technology-based approach to improving transparency and information accessibility,

rather than a system that automatically guarantees higher profits or eliminates all intermediaries.

The proposed architecture also provides scope for future improvements, including real-time market data integration, wider IoT monitoring, mobile accessibility and advanced agricultural analytics.

7.3 Discussion

One of the important aspects of AgriTrace is that it combines multiple technologies to address different problems in the agricultural supply chain.

The marketplace focuses on connecting farmers and sellers. The market-price module is intended to help farmers compare available prices across different locations. Blockchain provides a method for verifying selected records, while QR codes make batch information easier to access.

The proposed IoT integration adds another layer by connecting physical storage and transportation conditions with digital records.

However, the effectiveness of these features depends on the availability and reliability of the underlying data. For example, market price comparisons require authentic government or market records with clear observation dates. Similarly, sensor-based monitoring requires properly calibrated sensors and reliable communication.

Therefore, AgriTrace should be evaluated not only by its user interface but also by the accuracy, reliability and completeness of its integrated services.

7.4 Limitations

The following limitations need to be considered during the current development stage:

- Official agricultural data availability may differ between crops and districts.

- Market prices may be delayed or unavailable for certain markets.

- IoT monitoring requires physical hardware integration and testing.

- Sensor readings may be affected by placement, calibration and environmental conditions.

- Blockchain transactions may involve network fees and confirmation delays.

- Internet connectivity is necessary for most online operations.

- Accurate crop shelf-life prediction requires additional scientific data and validation.

These limitations provide opportunities for further development rather than reducing the overall value of the project.

CHAPTER VIII

COMMUNITY IMPACT

8.1 Impact on Farmers

AgriTrace is primarily designed to benefit farmers by improving their access to market information and potential buyers.

Through the digital marketplace, farmers can display their available produce and interact with sellers. District-wise market price comparison can also help them understand the available selling opportunities in different markets.

This can support better decision-making and reduce dependence on informal information sources.

8.2 Impact on Sellers and Buyers

Sellers can use the platform to discover agricultural products, compare available listings and interact directly with farmers.

Buyers can also benefit from improved access to product information and batch-level traceability.

This creates an opportunity for better communication and more transparent agricultural transactions.

8.3 Impact on Consumers

Consumers increasingly want to understand where their food comes from and how it has been handled.

AgriTrace aims to address this requirement through QR-based product identification and traceability.

By scanning a batch QR code, consumers can access the available product journey information and relevant verification details.

8.4 Reduction of Information Gaps

One of the major community benefits of AgriTrace is its focus on reducing information gaps between different participants in the agricultural supply chain.

Instead of relying entirely on disconnected communication methods, farmers, sellers and consumers can access relevant information through a common digital platform.8.5 Support for Sustainable Agriculture

The proposed IoT monitoring system can help improve awareness of storage and transportation conditions.

Monitoring temperature and humidity may support better handling practices and help identify conditions that require attention.

Better traceability and supply-chain visibility may also contribute to reducing avoidable losses, although actual reductions would need to be measured through field trials.

8.6 Sustainable Development Goals

AgriTrace has potential alignment with the following United Nations Sustainable Development Goals:

- SDG 2 – Zero Hunger: Supports agricultural market access and improved food supply-chain information.

- SDG 8 – Decent Work and Economic Growth: Encourages better market participation and transparent transactions.

- SDG 9 – Industry, Innovation and Infrastructure: Applies digital technologies, blockchain and IoT to agriculture.

- SDG 12 – Responsible Consumption and Production: Supports product traceability and improved supply-chain visibility.

- SDG 17 – Partnerships for the Goals: Connects farmers, sellers, buyers and other supply-chain participants.

Overall, AgriTrace aims to use technology to create practical benefits for agricultural communities and encourage a more connected agricultural ecosystem.

CHAPTER IX

CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

AgriTrace is proposed as a digital solution to address important challenges in the agricultural supply chain, particularly the lack of transparent market information, limited direct communication between farmers and sellers, and difficulties in verifying agricultural product information.

By combining a farmer–seller marketplace, district-wise price discovery, blockchain-based traceability and IoT-enabled monitoring, the platform aims to create a more connected and transparent farm-to-market system.

The project brings together modern web development technologies and emerging technologies to address practical agricultural requirements.

The marketplace provides a common space for product discovery and transactions. The traceability module supports the recording of product information, while blockchain technology can help verify the integrity of selected records. QR codes make this information accessible to buyers and consumers.

The proposed IoT integration can further improve visibility into storage and transportation conditions.

AgriTrace is not simply intended to be an online agricultural marketplace. Its broader purpose is to improve trust, information accessibility and communication throughout the agricultural supply chain.

9.2 Future Scope

AgriTrace can be expanded in several directions to make it more useful for farmers and other agricultural stakeholders.

1. Advanced Agricultural Price Prediction

Machine learning models can be introduced to analyse historical market prices, seasonal patterns and market arrivals. This may help estimate possible future price trends, provided sufficient reliable data is available.

2. Direct Integration with Government Agricultural Data

The platform can be connected with official Tamil Nadu Government, AGMARKNET and data.gov.in services wherever suitable APIs or permitted datasets are available.

This can provide more comprehensive crop and district-wise market information.

3. Advanced IoT Monitoring

Additional sensors can be integrated to monitor:

- Temperature.

- Humidity.

- Storage conditions.

- Location during transportation.

- Other relevant environmental parameters.

4. Crop-Specific Shelf-Life Estimation

Future versions can include crop-specific shelf-life estimation using scientific storage data, environmental readings and validated prediction models.

5. Mobile Application

A dedicated Android application can make AgriTrace more accessible to farmers who primarily use smartphones.

6. Regional Language and Voice Assistance

The platform can be improved with complete Tamil and other regional-language support, voice-based navigation and speech-to-text features.

This can make the application easier to use for people who are less comfortable with English-based digital platforms.

7. Blockchain-Based Digital Certificates

Verified agricultural batch information can be extended into digital certificates containing relevant origin, harvest and handling records.

8. Logistics and Transportation Integration

Future versions can connect farmers, sellers and transportation providers to coordinate product delivery and improve supply-chain visibility.

9. Farmer Analytics Dashboard

Advanced dashboards can provide information about:

- Product sales history.

- Order completion.

- Market price trends.

- Revenue summaries.

- Demand patterns.

- Reward-point history.

10. Expansion Across India

After successful implementation and testing in Tamil Nadu, AgriTrace can be extended to other Indian states by integrating their respective agricultural markets and government data sources.

9.3 Final Statement

The future of agriculture depends not only on increasing production but also on improving how agricultural products are marketed, transported, verified and delivered.

AgriTrace aims to contribute to this transformation by connecting farmers with markets through transparent digital processes.

"From Farm to Market, Building Trust Through Technology."

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INTERNAL QUALITY ASSURANCE CELL

PROJECT AUDIT REPORT

This is to certify that the project work entitled "Blockchain-Based Farm-to-Market Traceability System" categorized as an internal project done by **GURU PRASANTH S – 9925009028, LEUN HAMMI D - 99250040308, JEYAPRASANNA M - 99250040310** of the Department of Computer Science and Engineering, under the guidance of **GUIDE NAME** during the Even semester of the academic year 2025 - 2026 are as per the quality guidelines specified by IQAC.

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